]Mayur Shivamurthy
Email: mayur.sm.18@iitdh.ac.in

Design of Bandgap Voltage Reference Circuits.

CTAT Voltage Generation Circuit

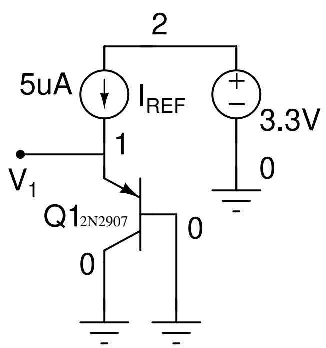

Figure 1: CTAT Generation Circuit

It can be observed that the voltage V_1 decreases with the increase in temperature T. Identify the slope $\partial V_1/\partial T$ for the above circuit.

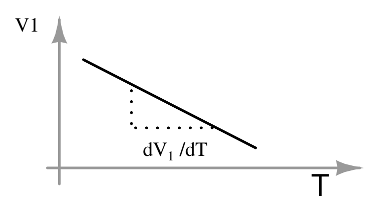

Figure 2: Variation of V_1 with temperature T

PTAT Voltage Generation Circuit

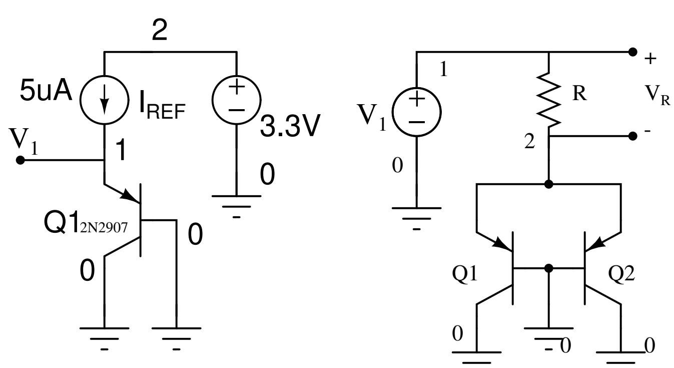

Figure 3: Basic PTAT Generation Circuit

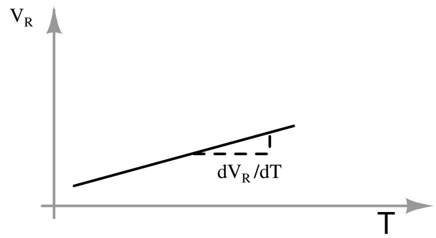

Figure 4: Variation of V_R with temperature T

It can be observed that the voltage V_R increases with the increase in temperature T. Identify the slope $\partial V_R/\partial T$ for the above circuit. Fig. 5 shows the PTAT Voltage generation with MOS current mirror.

$$V_R = V_T ln(N) \tag{1}$$

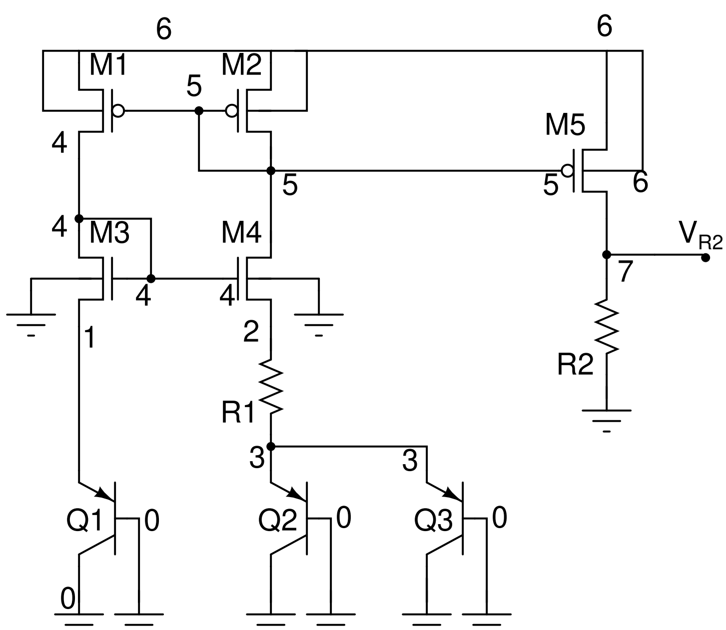

Figure 5: PTAT Voltage generation with MOS current mirror.

Design Exercise:

Design Bandgap voltage Reference circuit with $I_0 = 5\mu A$, $V_{DD} = 3.3$ V and N=2.

From the Fig. 6, V_{REF} is expressed as,

$$V_{REF} = V_D + V_{R2} = V_D + (R_2/R_1)ln(N)V_T \tag{2}$$

$$V_{REF} = \alpha_1 V_D + \alpha_2 V_T \tag{3}$$

On comparing Eq. 2 and Eq. 3

$\alpha_1 = 1$, $\alpha_2 = (R_2/R_1)ln(N)$.

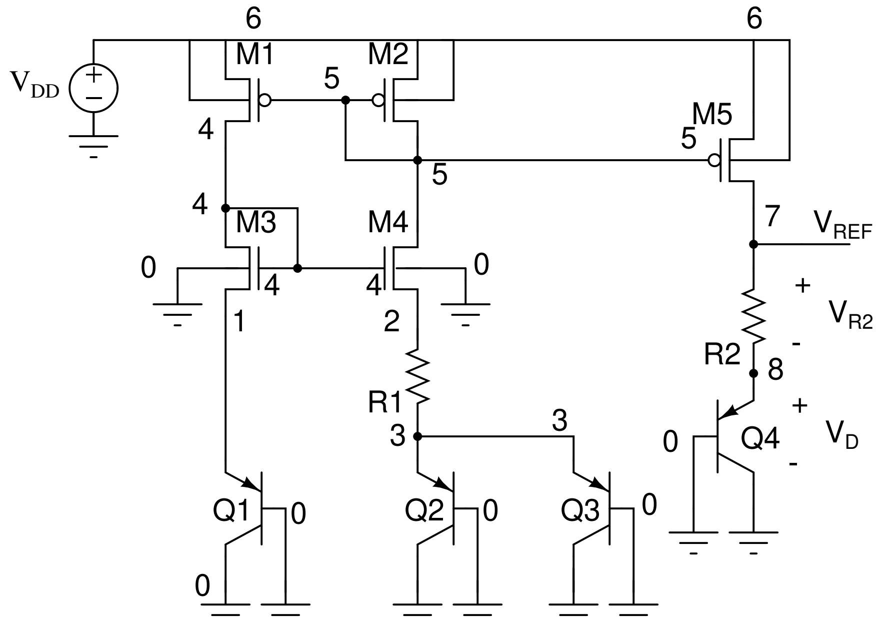

Figure 6: Voltage reference circuit with CTAT and PTAT combined.

For ZTAT behaviour of V_{REF}, it required to have

$$\frac{\partial V_{REF}}{\partial T} = \alpha_1 \frac{\partial V_D}{\partial T} + \alpha_2 \frac{\partial V_{R2}}{\partial T} = 0 \tag{4}$$

With $\alpha_1 = 1$, $\partial V_D/\partial T = -1.6mV/K$, $\partial V_T/\partial T = 86\mu V/K$, N=2.

α_2 is evaluated to 17.44 and hence the relation between R_1 and R_2 is written as,

$$R_2 = 25.163 R_1 \tag{5}$$

Evaluation of R_1

The voltage across R_1, V_{R1} is expressed as,

$$V_{R1} = I_o R_1 = V_T ln(N) \tag{6}$$

With $I_o = 5\mu A$, $V_T = 26\ mV$ [at T=27°C]$, N = 2$.

R_1 is evaluated to 3.6 $K\Omega$.

Evaluation of R_2

From Eq. 5, R_2 is evaluated to 90.5 $K\Omega$.

Exercise

1. Plot the voltages V_{R2} , V_D and V_{REF}. Observe the behaviour of these voltages with the variation in temperature.

2. Repeat 1 for different values of R_1, R_2 and N.

3. Observe the behaviour of circuit with the variation in supply voltage V_{DD}.

4. Simulate the voltage reference circuit as shown in Fig. 7.Use previously designed opamp for the simulation.

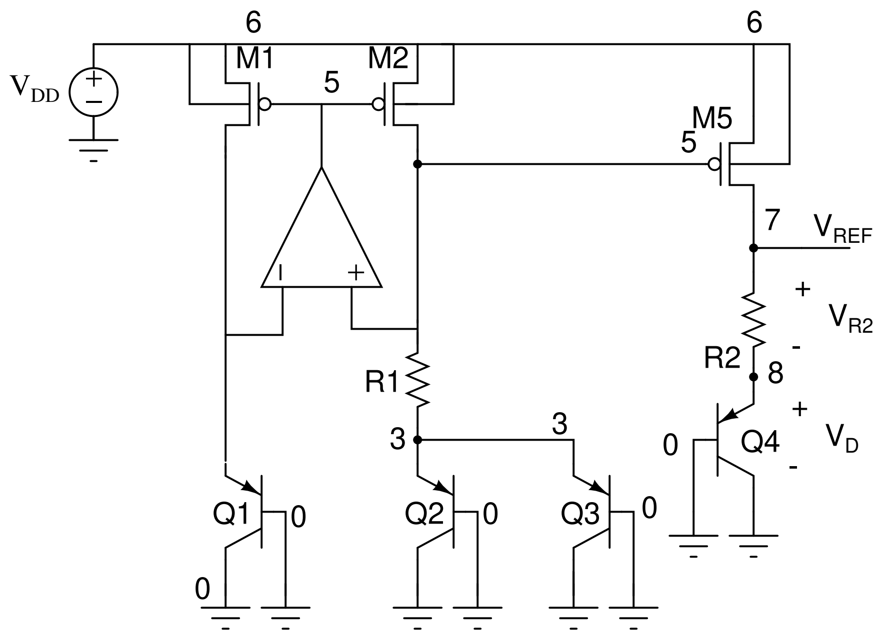

Figure 7: Voltage reference circuit with OPAMP.